

EXP 11- CREDIT SCORE CLASSIFICATION

NAME: SAMUEL I

REG NO :192312584

AIM

To develop a machine learning program that classifies individuals into different **credit score categories** based on financial and personal attributes such as income, loans, payment history, and credit utilization.

PROCEDURE

1. **Collect the Dataset:** Load a credit score dataset containing customer financial information and the corresponding credit score category.
2. **Preprocess the Data:** Handle missing values, remove irrelevant columns, encode categorical features, and normalize numerical features if required.
3. **Split the Dataset:** Divide the dataset into **training and testing sets** to train and evaluate the classification model.
4. **Train the Model:** Apply a suitable classification algorithm such as **Random Forest, Decision Tree, or Logistic Regression** to predict the credit score category.
5. **Evaluate the Model:** Predict the credit scores for the test data and evaluate the model using **accuracy, confusion matrix, precision, recall, and F1-score**.

CODE :

```
(l) import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix

data = {
    'Income':[25000,40000,55000,70000,30000,80000,45000,60000],
    'Loan':[15000,10000,8000,5000,18000,4000,12000,7000],
```

```
'PaymentHistory':[60,75,85,95,55,98,80,90],  
'CreditUtilization':[80,60,45,25,85,20,55,35],  
'Score':['Poor','Fair','Good','Excellent','Poor','Excellent','Fair','Good']  
}
```

```
df = pd.DataFrame(data)
```

```
X = df.drop('Score', axis=1)
```

```
y = LabelEncoder().fit_transform(df['Score'])
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.25, random_state=42  
)
```

```
model = RandomForestClassifier(n_estimators=100, random_state=42)  
model.fit(X_train, y_train)
```

```
pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, pred))
```

```
print("Confusion Matrix:\n", confusion_matrix(y_test, pred))
```

OUTPUT:

```
Accuracy: 1.0  
Confusion Matrix:  
[[1 0]  
 [0 1]]
```

```

(II) import pandas as pd

from sklearn.tree import DecisionTreeClassifier

from sklearn.preprocessing import LabelEncoder

data = {
    'Age':[21,25,30,35,40,45,28,50,32,38],
    'Savings':[1000,3000,8000,15000,25000,40000,5000,50000,12000,20000],
    'Number_of_Loans':[5,4,3,2,1,0,4,0,2,1],
    'Late_Payments':[7,6,4,2,1,0,5,0,2,1],
    'Score':['Poor','Poor','Fair','Good','Good',
            'Excellent','Poor','Excellent','Good','Excellent']
}

df = pd.DataFrame(data)

X = df.drop('Score', axis=1)

le = LabelEncoder()

y = le.fit_transform(df['Score'])

model = DecisionTreeClassifier(random_state=42)

model.fit(X, y)

new_customer = [[33, 14000, 2, 2]]

prediction = model.predict(new_customer)

print("Predicted Credit Score:", le.inverse_transform(prediction)[0])

```

OUTPUT:

```
Predicted Credit Score: Good
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(
```

(III)

```
import pandas as pd
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {
    'Card_Usage':[95,85,70,55,40,20,90,60,35,15],
    'Credit_Limit':[20000,25000,40000,50000,70000,100000,30000,60000,80000,120000],
    'Late_Payments':[8,6,4,3,1,0,7,2,1,0],
    'Years_of_Credit':[1,2,3,4,6,10,1,5,7,12],
    'Score':['Poor','Poor','Fair','Fair','Good',
            'Excellent','Poor','Good','Good','Excellent']
}
```

```
df = pd.DataFrame(data)
```

```
X = df.drop('Score', axis=1)
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Score'])
```

```
model = DecisionTreeClassifier(random_state=42)
```

```
model.fit(X, y)
```

```
new_customer = [[45, 75000, 1, 7]]
```

```
prediction = model.predict(new_customer)
```

```
print("Predicted Credit Score:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
Predicted Credit Score: Good
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(
```

(IV) CODE

```
import pandas as pd
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {
    'Job_Experience':[0,1,2,5,8,12,1,6,10,15],
    'Monthly_Income':[15000,20000,28000,40000,55000,80000,22000,45000,60000,100000],
    'Monthly_Expenses':[14000,19000,25000,30000,35000,40000,20000,32000,38000,45000],
    'Savings':[500,1000,3000,10000,20000,40000,1500,12000,25000,60000],
    'Score':['Poor','Poor','Fair','Fair','Good',
            'Excellent','Poor','Good','Excellent','Excellent']
}
```

```
df = pd.DataFrame(data)
```

```
X = df.drop('Score', axis=1)
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Score'])
```

```
model = DecisionTreeClassifier(random_state=42)
```

```
model.fit(X, y)
```

```
new_customer = [[7, 50000, 35000, 18000]]
```

```
prediction = model.predict(new_customer)
```

```
print("Predicted Credit Score:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
Predicted Credit Score: Good
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(
```

(V)

CODE:

```
import pandas as pd
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {
```

```
    'Monthly_Transactions':[5,10,20,30,40,50,8,25,35,45],
```

```
    'Average_Balance':[1000,2000,5000,10000,20000,35000,1500,7000,15000,28000],
```

```
    'Bounced_Cheques':[5,4,3,2,1,0,4,2,1,0],
```

```
    'Account_Age':[1,2,3,5,7,10,1,4,6,9],
```

```
    'Score':['Poor','Poor','Fair','Fair','Good',
```

```
            'Excellent','Poor','Good','Good','Excellent']
```

```
}
```

```
df = pd.DataFrame(data)
```

```
X = df.drop('Score', axis=1)
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Score'])
```

```
model = DecisionTreeClassifier(random_state=42)
```

```
model.fit(X, y)
```

```
new_customer = [[32, 9000, 1, 5]]
```

```
prediction = model.predict(new_customer)
```

```
print("Predicted Credit Score:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
Predicted Credit Score: Fair
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(
```